

Random Variables

A random variable takes numerical values that describe the outcomes of some chance process. Random variables involve the same probability rules we have already learned, but we will extend those rules to be able to model more events.

Example: Suppose we toss a coin 3 times.

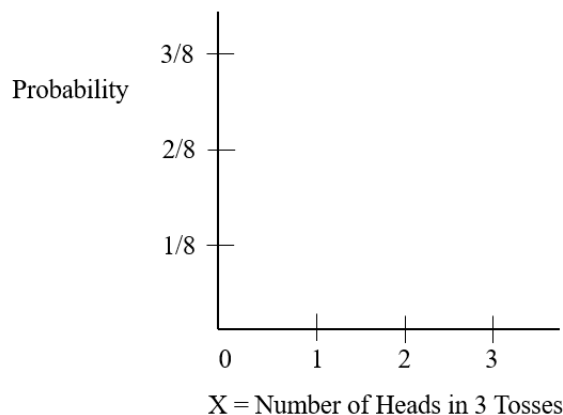
The sample space for this chance process is:	Total Number of Outcomes:
	Probability of any one outcome:

Let X = Number of Heads out of 3 coin tosses

Symbols	Meaning	Probability
$X = 0$		
$X = 1$		
$X = 2$		
$X = 3$		

With these 4 X values, and their corresponding probabilities, we can create a frequency table and a histogram to describe this event.

X	0	1	2	3
P(X)				



- The frequency table and histogram above each represent a _____.
- A probability distribution tells us the value that our random variable can take and the probability associated with each value.

Probability Distributions: Requirements

Every probability distribution must satisfy each of the following requirements.

1. There is a numerical (not categorical) _____, and each of the values of X has an associated probability with it.
2. The sum of all the probabilities in the distribution _____ must equal _____
3. _____ for all probabilities in the distribution.

Notation:

Notation	What it means	Example					
$P(X = 2)$	What is the probability that the random variable equals 2?	$P(X = 2)$					
$P(X \geq 2)$	What is the probability that the random variable is greater than or equal to 2?	$P(X \geq 2)$					
$P(X < 1)$	What is the probability that the random variable is less than 1?	$P(X < 1)$					

X	0	1	2	3
P(X)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Discrete Random Variable

A discrete random variable takes on a fixed set of possible values with whole number outcomes.

- Random variables are usually capital letters (X , Y , Z , etc.)
- Random variables can be discrete or continuous (continuous is next lesson)
- Random variables must be numeric in value
 - In our example, even though a result of Head or Tails is categorical, we described it numerically.

The mean of a discrete random variable X , is the mean outcome for _____.

We think of this as the “expected value” because it is the average value we would expect to get if the trials could continue indefinitely. (Not what we expect to get in a single trial)

To find the mean, _____, or expected value, _____, of a discrete random variable X , multiply each possible value by its probability, then add all the products.

					$\mu_x =$
X					
P(X)					$\mu_x =$

Example: Northwestern University posts the grade distribution for its courses online. Students in Statistics 101 in a recent semester received 26% A's, 42% B's, 20% C's, 10% D's, and 2% F's.

a) Create a probability distribution for the random variable X , which represents the student's grade on a four point scale (with A = 4).

b) Find and interpret $P(X \geq 3)$ if X represents a randomly selected Statistics 101 student.

c) Write the event “the student got a grade worse than C” in terms of values of the random variable X . What is the probability of this event?

d) Compute the mean of the random variable X . Interpret this value in context.

Standard Deviation of a Random Variable

The standard deviation of a random variable X is a measure of how much the values of the variable typically vary from the expected value.

<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 15%; padding: 5px;">X</td> <td style="width: 15%; height: 25px;"></td> <td style="width: 15%; height: 25px;"></td> <td style="width: 15%; height: 25px;"></td> <td style="width: 15%; height: 25px;"></td> </tr> <tr> <td style="padding: 5px;">P(X)</td> <td style="height: 25px;"></td> <td style="height: 25px;"></td> <td style="height: 25px;"></td> <td style="height: 25px;"></td> </tr> </table>	X					P(X)					$\sigma_x^2 =$ $\sigma_x^2 = Var(X) =$ $\sigma_x =$
X											
P(X)											

Example: For the Northwestern example above, find and interpret the standard deviation of the grade probability distribution.

Continuous Random Variable

- Continuous Random Variables take on all possible values in an _____ of numbers. The probability distribution of X is described by a _____.
- The probability of any event is the area under the density curve and above, below, or between the values x that define the event.
- Area under a density curve is always equal to _____
- The probability is on the y-axis in a distribution, so finding the area equates to finding the _____ of getting an interval of numbers.
- The probability _____ (for example, $P(X = 2)$) is _____ for a continuous random variable because the area under a point is 0 (this ties into the calculus of statistics).
- The mean and standard deviation of a continuous random variable are found using integration, so we will not discuss their formulas here.

Continuous Random Variables are usually represented as functions in practice. The continuous random variable we will explore is the Normal Distribution, and we will revisit it at the start of Unit 5.

Why do we have random variables?

This is an important question to keep in mind as we continue to look at different random variables. We need random variables because they can represent more things that a single measurement can. One-variable statistics used data, where random variables can use theory and apply it to real situations.